

-1	0	+1
-2	0	+2
-1	0	+1

G_x

-1	-2	-1
0	0	0
+1	+2	+1

G_y

$$G = \sqrt{G_x^2 + G_y^2}, \quad \theta = \arctan(G_y/G_x)$$

Sobel (1968)